

```
# Day 4 - Pandas Data Cleaning and Analysis

import pandas as pd

# Create sample lead data with missing and duplicate values
data = {
    "Name": [
        "Rahul", "Priya", "Amit", "Sneha", "Rohan",
        "Neha", "Vikas", "Anjali", "Karan", "Pooja",
        "Rahul", "Priya", "Arjun", "Meena", "Sahil"
    ],
    "Category": [
        "AI", "Web Development", "AI", "Data Analysis", "Web Development",
        "AI", "Data Analysis", "AI", "Web Development", "Data Analysis",
        "AI", "Web Development", "AI", "Data Analysis", "Web Development"
    ],
    "Value": [
        500, 700, None, 450, 800,
        600, 550, None, 900, 400,
        500, 700, 750, None, 650
    ],
    "Status": [
        "Interested", "Interested", "Contacted", "Interested", "Converted",
        "Contacted", "Interested", "Converted", "Contacted", "Interested",
        "Interested", "Interested", "Converted", "Contacted", "Interested"
    ]
}

# Create a DataFrame
df = pd.DataFrame(data)

# Save the original data as a CSV file
df.to_csv("leads.csv", index=False)

print("Original Dataset:")
display(df)
```

Original Dataset:

	Name	Category	Value	Status
0	Rahul	AI	500.0	Interested
1	Priya	Web Development	700.0	Interested
2	Amit	AI	NaN	Contacted
3	Sneha	Data Analysis	450.0	Interested
4	Rohan	Web Development	800.0	Converted
5	Neha	AI	600.0	Contacted
6	Vikas	Data Analysis	550.0	Interested
7	Anjali	AI	NaN	Converted
8	Karan	Web Development	900.0	Contacted
9	Pooja	Data Analysis	400.0	Interested
10	Rahul	AI	500.0	Interested
11	Priya	Web Development	700.0	Interested
12	Arjun	AI	750.0	Converted
13	Meena	Data Analysis	NaN	Contacted
14	Sahil	Web Development	650.0	Interested

```
# Load the CSV file
df = pd.read_csv("leads.csv")

# Display the first few rows
print("First 5 rows:")
display(df.head())

# Check the dataset information
print("\nDataset Information:")
df.info()

# Check for missing values
print("\nMissing Values:")
print(df.isnull().sum())
```

First 5 rows:

	Name	Category	Value	Status
0	Rahul	AI	500.0	Interested
1	Priya	Web Development	700.0	Interested
2	Amit	AI	NaN	Contacted
3	Sneha	Data Analysis	450.0	Interested
4	Rohan	Web Development	800.0	Converted

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 15 entries, 0 to 14

Data columns (total 4 columns):

#	Column	Non-Null Count	Dtype
0	Name	15 non-null	object
1	Category	15 non-null	object
2	Value	12 non-null	float64
3	Status	15 non-null	object

dtypes: float64(1), object(3)

memory usage: 612.0+ bytes

Missing Values:

Name	0
Category	0
Value	3
Status	0

dtype: int64

```
# Check the number of duplicate rows
print("Duplicate rows before cleaning:", df.duplicated().sum())

# Remove duplicate rows
df = df.drop_duplicates()

# Fill missing values in the Value column with the average value
average_value = df["Value"].mean()
df["Value"] = df["Value"].fillna(average_value)

# Standardize the Status column
df["Status"] = df["Status"].str.strip().str.title()

print("\nDataset after cleaning:")
display(df)

# Check for missing values after cleaning
print("\nMissing values after cleaning:")
print(df.isnull().sum())

# Check duplicates after cleaning
print("\nDuplicate rows after cleaning:", df.duplicated().sum())
```

Duplicate rows before cleaning: 2

Dataset after cleaning:

	Name	Category	Value	Status
0	Rahul	AI	500.0	Interested
1	Priya	Web Development	700.0	Interested
2	Amit	AI	630.0	Contacted
3	Sneha	Data Analysis	450.0	Interested
4	Rohan	Web Development	800.0	Converted
5	Neha	AI	600.0	Contacted
6	Vikas	Data Analysis	550.0	Interested
7	Anjali	AI	630.0	Converted
8	Karan	Web Development	900.0	Contacted
9	Pooja	Data Analysis	400.0	Interested
12	Arjun	AI	750.0	Converted
13	Meena	Data Analysis	630.0	Contacted
14	Sahil	Web Development	650.0	Interested

Missing values after cleaning:

```
Name      0
Category   0
Value      0
Status     0
dtype: int64
```

Duplicate rows after cleaning: 0

```
# Filter leads whose Value is greater than 600
high_value_leads = df[df["Value"] > 600]
```

```
print("Leads with Value greater than 600:")
display(high_value_leads)
```

Leads with Value greater than 600:

	Name	Category	Value	Status
1	Priya	Web Development	700.0	Interested
2	Amit	AI	630.0	Contacted
4	Rohan	Web Development	800.0	Converted
7	Anjali	AI	630.0	Converted
8	Karan	Web Development	900.0	Contacted
12	Arjun	AI	750.0	Converted
13	Meena	Data Analysis	630.0	Contacted
14	Sahil	Web Development	650.0	Interested

```
# Calculate average Value for each Category
category_summary = df.groupby("Category")["Value"].mean()
```

```
print("Average Value by Category:")
display(category_summary)
```

Average Value by Category:

Value	
Category	
AI	622.0
Data Analysis	507.5
Web Development	762.5

`dtype: float64`

Start coding or [generate](#) with AI.

Day 4 — Pandas Explanation

Pandas is a Python library used to work with tabular data.

What I Practiced

- Creating a DataFrame
- Creating and loading a CSV file
- Inspecting data
- Detecting missing values
- Removing duplicate rows